

Multiple Choice (10 marks)

1. The emission spectrum of the doubly ionized lithium atom Li^{++} ($Z = 3$, $A = 7$) is identical to that of a hydrogen atom in which all the wavelengths are
 - (a) decreased by a factor of 9
 - (b) decreased by a factor of 49
 - (c) decreased by a factor of 81
 - (d) increased by a factor of 9
2. The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is
 - (a) $1.2 \times 10^9 \text{ m/s}$
 - (b) $3.0 \times 10^8 \text{ m/s}$
 - (c) $1.5 \times 10^8 \text{ m/s}$
 - (d) $1.0 \times 10^8 \text{ m/s}$
3. If the total energy of a particle of mass m is equal to twice its rest energy, then the magnitude of the particle's relativistic momentum is
 - (a) $mc/2$
 - (b) $mc/(2^{1/2})$
 - (c) mc
 - (d) $(3^{1/2})mc$
4. An organ pipe, closed at one end and open at the other, is designed to have a fundamental frequency of C (131 Hz). What is the frequency of the next higher harmonic for this pipe?
 - (a) 44 Hz
 - (b) 196 Hz
 - (c) 262 Hz
 - (d) 393 Hz
5. The best type of laser with which to do spectroscopy over a range of visible wavelengths is
 - (a) a dye laser
 - (b) a helium-neon laser
 - (c) an excimer laser
 - (d) a ruby laser

True / False (10 marks)

Answer True or False

6. True or False: Electromagnetic radiation emitted from a nucleus is most likely in the form of gamma rays.
7. True or False: The electric displacement current through a surface S is proportional to the rate of change of the electric flux through S.
8. True or False: The observer measures the meter stick passing in 1.6 ns due to its high speed of 0.8 c.
9. True or False: The electric displacement current through a surface S is proportional to the rate of change of the electric flux through S.
10. True or False: An observer must move at 0.60 c parallel to a 1.00 m rod to measure its length as 0.80 m due to length contraction.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In a Michelson interferometer with a gas cell of 10 cm optical path length, 100 fringes are counted during evacuation. Calculate the original gas's index of refraction and discuss the implications of your findings.
12. Discuss the role of dopants in germanium semiconductors, specifically identifying which ions cannot be used for n-type doping and explaining the underlying reasons for their incompatibility.

End of Paper